

Real-time Tracking and Localization of Rapid Instantaneous SOP Fluctuations for 100G Coherent System over FPGA

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Abstract We demonstrate a two-step AEQ cooperating a loopback configuration to track and locate rapid instantaneous state of polarization in a 100G 16QAM coherent optical system based on FPGA. The localization error is within 115 m at 5 MHz. ©2026 The Author(s)

Introduction

Polarization fluctuation is a critical phenomenon in optical fiber system maintenance, enabling applications such as co-trench/cable detection, lightning strike localization, and proactive fiber break prevention [1-4]. Various strategies have been proposed to monitor and track the state of polarization (SOP) changes in optical links. For instance, pilot and symbol based methods geometrically analyze the SOP of periodic pilot symbols in Stokes space to estimate the rotation axis and angle, achieving polarization change monitoring from quasi-static to 3 MHz [1]. Alternatively, optical supervisory channel (OSC) based schemes employ a polarization beam splitter (PBS) and two detectors to capture polarization transients per span; when configured in bidirectional transmission, they can localize instantaneous SOP fluctuations with an accuracy of less than 300 m [2,3]. For ultra-fast SOP rotation, chirp-pilot signals processed through fractional Fourier transform (FrFT) have been demonstrated to track rotation speed without sacrificing spectral efficiency [4]. However, a simple method that does not alter the existing optical network structure for SOP sensing seems more attractive.

In recent years, DSP has been a hot topic, offering an integrated and cost-effective solution for optical sensing and maintenance. In a coherent receiver, the adaptive equalizer (AEQ) within the DSP module can compensate for

inter-symbol interference (ISI) as well as perform polarization demultiplexing and compensate for polarization mode dispersion (PMD). The AEQ-dependent method directly utilizes the tap coefficients output by the AEQ during its polarization demultiplexing operation to retrieve the dynamic rotation characteristics of the SOP. This method doesn't need additional devices such as PBS or additional pilot symbols, which makes the system simple, cost-effective, and more importantly, compatible with existing optical transport network (OTN) equipment architectures. Cooperating with synchronization technology, the AEQ-dependent method becomes a promising approach for detecting and localizing fiber events in OTN.

In this work, we employ a two-step AEQ and loopback configuration for tracking and localization of rapid instantaneous SOP fluctuations in a 100G 16QAM optical system over FPGA. SOP fluctuations tracking and localization under different rotation frequency and different fiber length are investigated.

SOP Tracking and Localization

Schematic diagram of the two-stage AEQ for SOP tracking is shown in Fig. 1(a). A 2×2 AEQ is usually employed in classic coherent transceiver oDSP flow for polarization demultiplexing and compensating for various linear impairments, including residual chromatic dispersion, polarization-dependent loss, and

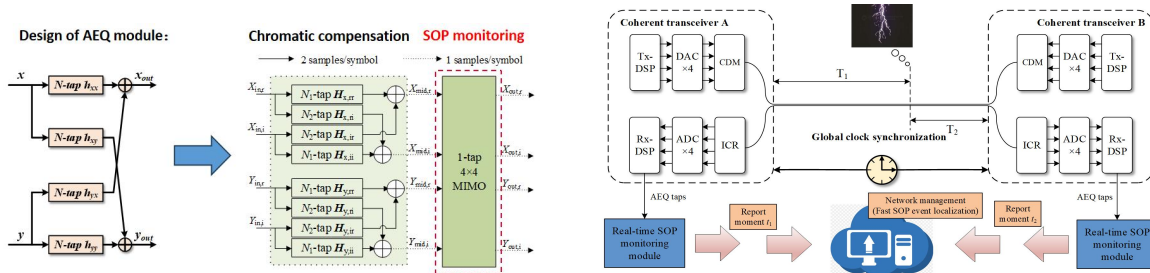


Fig. 1: (a) Schematic diagram of the two-stage AEQ. (b) Localization of rapid instantaneous SOP fluctuations.

various ISI. To compensate for IQ impairments, a 4×4 structured AEQ as shown in Fig. 1(a) is employed. We improve the AEQ structure by using the first stage for residual chromatic dispersion compensation and the second stage (a single-tap 4×4 MIMO) for polarization demultiplexing. Due to the reduced number of taps, the convergence speed and the tracking speed for SOP rotations are enhanced, which facilitates real-time monitoring, localization, and alarming of high-speed SOP rotations. The Stokes vector of the signal can be calculated by directly extracting the AEQ filter tap coefficients which is the inverse of the Jones matrix of the optical fiber channel. Subsequently, the variation in the SOP can be determined from the change in the Stokes vector [5], thereby enabling monitoring of the signal SOP rotation speed.

In OTN systems, localization based on time difference is a feasible approach. In this work, a loopback configuration is adopted to achieve fiber event localization using time difference method. The principle is illustrated in Fig. 1(b). The distance between coherent transceiver A and B is L . The location L_1 of fiber event can be obtained through the time difference ΔT between the arrivals of the event at the two receivers, which can be expressed as

$$L_1 = (L - \Delta L)/2 \quad (1)$$

where $\Delta L = c|\Delta T|$ and c is the speed of light in optical fibers.

Experiment Demonstrations

Figure 2 shows the SOP tracking and localization testbed of a 100G dual-polarization 16QAM optical system over FPGA. Xilinx Virtex UltraScale+ XCVU13P FPGA is employed. As shown in Fig. 2(a), in the transmitter side, by utilizing parallel PRBS generator, Gray DP-AQM modulation and FIR, together with 4 Digital-to-Analog Converters (DACs), 4 branches of

14GBaud data rate signal are generated to drive the Coherent Driver Modulator (CDM). Spectrum of the driven signal is shown in Fig. 2(b). To simplify system, a loopback configuration is adopted to locate fast polarization rotation events. The amplified signal injects into port 1 of the first optical circulator (OC₁). Two polarization scramblers (EPS 1000) are inserted in the optical fiber link. The scramblers can generate high-speed SOP rotation to simulate disturbance events. The OC₂ is used to loop back the transmitted signal. The optical signal returns to port 2 of OC₁ and is then sent to the receiver side via port 3.

In the receiver side, signal is received in a integrated coherent receiver (ICR). The 4 branch signals after Analog-to-Digital Converter (ADC) are led into the coherent oDSP board, where two stage-AEQ combining with static dispersion equalization, improved clock recovery based on Mueller and Müller, and carrier recover based on Viterbi-Viterbi, are used to enhance the ability to demodulate signal as well as track SOP. Highly reliable tap coefficients for subsequent monitoring modules is prepared for the SOP tracking module during the DSP process. The SOP tracking FPGA module is an external module that connects to the Rx-FPGA board via a 1.25Gb/s high-speed SFP optical module, extracting and analyzing the AEQ tap coefficients in real time to monitor the signal polarization state and to enable alarming and localization of fast polarization rotation events. The constellation diagram of the received signal after 86.597-km 2-span transmission is shown in Fig. 2(c) and the demodulated constellation diagram extracted from the receiving FPGA card shows bit error rate (BER) about 8.7×10^{-3} .

Experimental results

We performed experimental verification of the

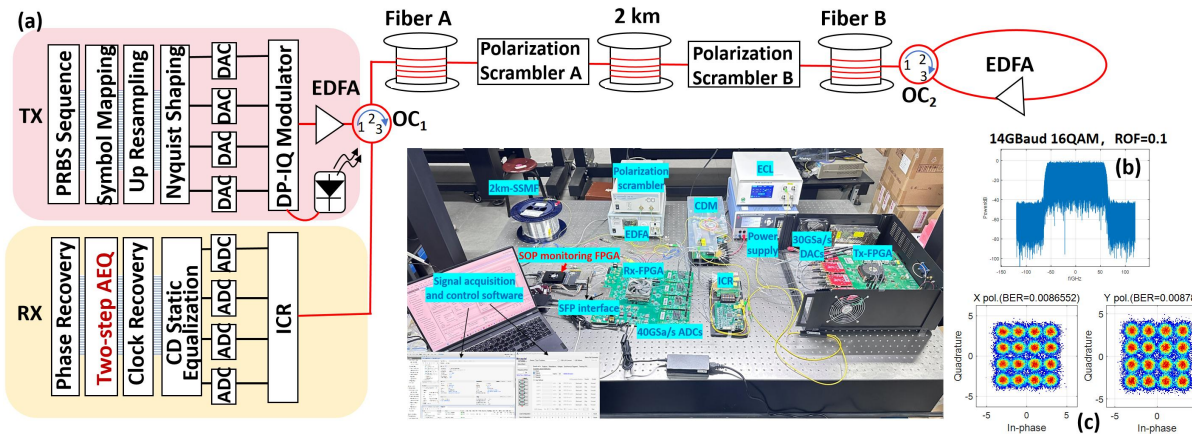


Fig. 2: (a) Experimental setup of the SOP tracking and localization based on FPGA. DAC: Digital-to-Analog Converter, OC: optical circulator, EDFA: Erbium-doped fiber amplifier, ICR: integrated coherent receiver, ADC: Analog-to-Digital Converter, (b) Spectrum of the driven signal, and (c) Constellation diagram of the received optical signal after 86.597-km transmission.

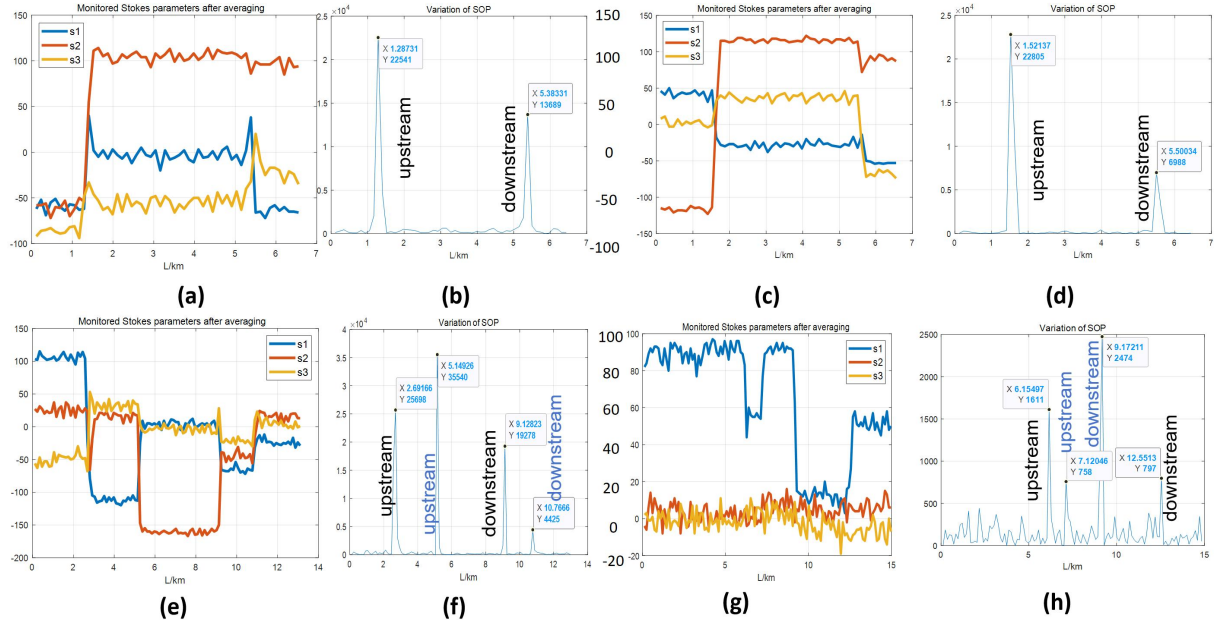


Fig. 3: Experimental results showing the Stokes parameters after averaging and the variation of SOP. (a)(b) SOP rotations at 5 Mrad/s with fiber length $L = 4.024$ km, $L_1 \approx 0$ km. (c)(d) SOP rotations at 1 Mrad/s with $L = 4.024$ km, $L_1 \approx 0$ km. (e)(f) Simultaneous SOP rotations at 5 Mrad/s and 1 Mrad/s with $L = 8.116$ km, $L_1 \approx 0$ km and 2.046 km, respectively. (g)(h) Simultaneous SOP rotations at 5 Mrad/s and 1 Mrad/s with $L = 86.597$ km, $L_1 \approx 40.153$ km and 42.2885 km, respectively.

proposed scheme. Firstly, the 100G signal was transmitted over a 4.024-km fiber link (fiber A and B were skipped). The polarization scrambler B was used to generate SOP rotations at 5 Mrad/s with a duration of 200 ns and a period of 35 μ s. The Stokes parameters of the signal, calculated from the collected data using the AEQ tap coefficients, are shown in Fig. 3(a). It can be seen that two obvious jumps occur in the 35- μ s monitoring time window, corresponding to two fast SOP rotations experienced by the upstream and downstream signals. By calculating the rate of change of SOP, two peaks were observed as shown in Fig. 3(b). The vertical axis represents the amplitude of the peak, i.e., the rate of change of SOP, but it is not exactly equal to the SOP rotation rate in the fiber. L_1 was calculated to be -0.036 km, corresponding to a theoretical value of approximately 0. To verify the effectiveness of the system for different rotation rates, we set the rotation rate to 1 Mrad/s. The test results are shown in Fig. 3(c) and (d). L_1 was calculated to be 0.023 km.

To investigate the tracking and localization of fiber events at different positions simultaneously, we used polarization scramblers A and B to apply perturbations at 5 Mrad/s and 1 Mrad/s, respectively. The length of fiber B is 2 km. The experimental results are shown in Fig. 3(e) and 3(f). The first and third peaks observed originate from the perturbation of polarization scrambler A, while the second and fourth peaks originate from the perturbation of polarization

scrambler B. The positions are calculated to be 0.021 km and 2.069 km, corresponding to theoretical values of 0 and 2.046 km, respectively. Similarly, polarization scramblers A and B both perturb at 5 Mrad/s with 86.6-km optical fiber link are tested as shown in Fig. 3(g) and 3(h). The L_1 is calculated to be 40.1 km and 42.273 km, corresponding to theoretical values of 40.153 and 42.288 km, respectively, with location error less than 115 m. The error mainly originates from factors such as sampling-induced errors, clock asynchronization of the polarization perturbation, and fiber measurement errors in the experiment.

Conclusions

We demonstrate a two-step AEQ with a loopback configuration for real-time tracking and localization of rapid SOP fluctuations in a 100G 16QAM FPGA system. The two-step AEQ uses a first stage for residual CD compensation and a second stage for polarization demultiplexing, accelerating convergence and tracking speed. Localization error within 115 m at 5 MHz is achieved. The method supports multi-span measurement, passes through amplifiers, and leverages integrated sensing and communication advantages. It is compatible with existing OTN architectures for engineering deployment.

Acknowledgements

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